

Lab 2: Data, Questions, and Variables

Choosing the dataset for Report 1, and drafting its card, its research question, and its variables of interest

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DELIVERABLES — DUE SUNDAY 20 SEPTEMBER, 11:59 PM, ON BRIGHTSPACE

- ☐ Lab-2_FIRSTNAME_LASTNAME.Rmd/Lab-2_FIRSTNAME_LASTNAME.qmd: the starter sheet, renamed, with the four parts completed and the generative AI disclosure filled in
- ☐ Lab-2_FIRSTNAME_LASTNAME.pdf: the same sheet, knitted to PDF
- ☐ the data file the sheet reads

This lab, like Lab 1, is done at home. It is released on Monday 14 September, and the completed sheet goes into the Lab 2 folder on Brightspace by Sunday night; there is no section meeting for it. (Remember to use the Brightspace page for your lecture section.) It has four parts, in the order the opening sections of the analysis reports: First, find a dataset and take a first look at it; second, write its dataset card; third, write a research question; fourth, list the variables of interest. In Lab 1, you wrote a card for a dataset we gave to you. The dataset you find this week comes with whatever information you can get from its website, which is the usual case.

Report 1 (exploratory data analysis and visualization) is set on Tuesday 22 September and due on Friday 9 October. This lab is a rehearsal for choosing its dataset, not the choice itself! The note at the end of Part 1 gives you more details on that.

about ten minutes to read, and about two hours in total

What to download and where to put it

Two files are on Brightspace for this lab:

- this handout;
- the starter sheet, lab-02-data-questions-and-variables.Rmd **OR** lab-02-data-questions-and-variables.qmd.

You will find a dataset in Part 1. Please make a folder for the lab inside the course

The .Rmd/.qmd and the data/ folder have to be saved together. The sheet opens the data by a relative path and cannot find it otherwise.

folder (labs/lab-02/ for instance), put the .Rmd/.qmd in it, and put the data file in a data/ folder beside the .Rmd/.qmd, as in Lab 1. Some of the sites in Part 1 ask for a free account before they let you download a file — Kaggle is one of those.

What to hand in, and when

Rename the starter to:

Lab-2_FIRSTNAME_LASTNAME.Rmd/lab-02-data-questions-and-variables.qmd

Work through it, knit it to PDF, and submit both files on Brightspace by the due date in the box above, together with the data file the sheet reads (the syllabus asks for the data whenever it is not a course file).

NOTE — HOW THIS LAB IS GRADED

Labs are out of 5 points. The syllabus rubric gives one point for arriving on time and one for a TA checking your progress at the end, and neither exists for a lab done at home. As for Lab 1, all five points come from the submission: the completed Rmd sheet and the three files uploaded, on time. A sheet with parts left blank is a partial submission and will be marked as such.

HINT

If knitting to PDF will not work: the syllabus's fallback applies: knit to HTML, convert the HTML to a PDF as shown in class, or with an online converter such as <https://www.sejda.com/html-to-pdf>, and submit that. Please also tell your TA, so that this process is sorted out before Report 1.

Part 1: Finding a dataset

Report 1's theme is exploratory analysis and visualization, so almost any table of a few hundred rows with a mix of numeric and categorical columns is a potentially viable choice. The checks below are for the things that cannot be fixed later.

Check	Why it matters
A few hundred rows or more	A distribution needs enough values to have a shape, and a comparison between groups needs enough rows in each group.
At least two numeric columns and one categorical column	So that a relationship between two quantities and a difference between groups are both possible.
A unit of observation you can name	“One row is one ...” has to have an answer (field 3 of the card).
Documentation	A page, a README, or a paper that says who collected the data and how. A file on its own cannot fill fields 2, 3, 4, and 9.
A licence	Stated somewhere on the page, and permitting use. “Unknown” is a finding for field 4, but a stated licence is better.
A size your laptop can open	Under 100 MB as a CSV is safe. Kaggle also hosts files of several gigabytes, and RStudio will not thank you for trying to open those up.

Two kinds of dataset are ruled out.

The first is anything that ships with R/Python or has been used in this course or is generally used for data science examples: `mtcars`, `iris`, the penguins, the yields, PluckData, anything on the UC Irvine Machine Learning Repository. Every question from those datasets have been answered many times over, and the card is usually trivial to write from the help page or from the course’s own pages.

The second is data from another course you have taken or are taking. (An exception: If you have research data that you have permission to analyze in this course, please feel free to do so. They make for the most interesting datasets you can look at.) Anything else is allowed; if you are unsure whether a dataset will carry a report, please email satadisha.saha.bhowmick@vanderbilt.edu with a link before you spend an evening on it.

Where datasets are found

There is no list of approved datasets. There are places where datasets are posted, and each of them holds thousands of potential datasets. The table gives seven, with what is there, whether a download needs an account, and what to expect by way of documentation.

On any site where users post files, the poster is rarely the person who collected the data. Who posted it is field 1 of the card; who collected it is field 2.

Site	What is there	Account, documentation, licence
Kaggle datasets	Files posted by users on every subject: sport, music, games, prices, health, climate. Search by keyword; filter by file type (CSV) and size.	A free account is needed to download. Documentation is the poster's description and the Discussion tab; the licence is a panel on the page and is sometimes "Unknown".
FRED	Economic time series, one per page, from the US and elsewhere: rates, prices, employment, output. Several series can be joined on their dates, as in the week 3 lectures.	No account for a CSV download. Every series page names its source, units and frequency. Public data from the agencies named on each page.
Bureau of Labor Statistics	US employment, unemployment, wages, prices (the CPI), hours, and time use, by industry, occupation, state and city.	No account; the data tools save a table as a spreadsheet or CSV. Documentation is the survey's methodology pages. US government data, in the public domain.
Open Science Framework	Data and materials that researchers deposit alongside their papers, across psychology, medicine, economics and the sciences.	No account to download. Documentation is whatever the project holds: a README, a codebook, a paper, sometimes nothing. The licence is chosen per project and shown on its page.
data.gov	The US federal open-data catalogue: hundreds of thousands of datasets from federal agencies, and many state and city portals.	No account. Documentation and licence vary by the publishing agency and are on the dataset's page.
Our World in Data	Country-by-year series on population, health, energy, education, and the like, each drawn as a chart with the data behind it.	No account; every chart has a download button for its CSV. Each page names the original source. Their own work is CC BY; the underlying data keep their sources' licences.
TidyTuesday	A dataset a week since 2018, chosen for exploration in R, each with a README naming its source and its variables.	No account; the CSV is on GitHub. Documentation is the README and the source it links to. The READMEs themselves are CC0; the data keep the original source's licence.

The **Data Resources** available on Brighspace lists a few more. Local portals are also worth a look: most cities and states run an open-data site, and Nashville's is at <https://data.nashville.gov>.

TASK 1 — FIND, DOWNLOAD, AND LOOK

Please find a dataset on one of the sites above (or another you know), download it, and put the CSV in your `data/` folder. Read it with `read_csv()` in the sheet's first chunk, and run the five commands from the Tukey lecture on it: `dim()`, `names()`, `glimpse()`, `summary()`, and `count()` on one categorical column. Then answer the checklist in the sheet, one line per check. If the dataset fails a check, please try another; an hour spent finding a dataset that passes is an hour well spent. Asking LLM agents about interesting datasets that they can help source and download is also a viable strategy.

NOTE — NOTHING IN THIS LAB IS BINDING

This lab is an opportunity to step through finding and describing a dataset, from the search box to a final dataset card, once, before it counts. The analysis report does not have to use the same dataset. It is your responsibility to find a dataset and a question that are rich enough to support an exploration, and that is a judgment you can only make by trying: writing the card and the question for a dataset is the way to find out

whether it will carry a report. If, by the end of Part 3, the dataset seems thin or the question narrow, or you find you are not interested in the answer, that is the best clue you will get that a report from that dataset won't be particularly interesting. Instead, go and find a dataset that is interesting enough. Changing datasets between this lab and the report costs you an evening; forcing a weak question through a whole report costs much more.

Part 2: The card for a posted dataset

The nine fields are those of Lab 1, Part 3. The table is repeated here so that it is in front of you.

Field	What it answers
1. Name and source	What the dataset is called, where you obtained it (a readable link), who posted it there, and the date you downloaded it.
2. Collector and purpose	Who originally collected the data and why. Often not the same as who posted it.
3. Collection	How, when, and where the data were gathered, and what one row represents (the unit of observation).
4. Licence and permissions	What you are allowed to do with the data. If the source says nothing, say that.
5. Structure	Rows and columns, and a table of the variables: name, meaning, the type R stored it as, units, range or levels, and a note where the stored type and the documentation disagree.
6. Missing values	How missingness is coded in the file, how many values are missing, in which columns, and whether there is a pattern.
7. Known issues	Caveats the documentation gives, and anything you found yourself that a reader should know.
8. Changes you made	Renamed columns, removed rows, converted types. Nothing, if nothing.
9. Citation	How the source asks to be cited, and the citation itself.

What is different this time

The penguins came with a help page, a package website, and three archive records. A dataset from one of the sites above comes with one page, written by whoever posted it. You should be upfront in your dataset card about the information you can find and what you can't find. Field by field:

1. Name and source.

The site, the page, who posted it, and the version. Kaggle datasets carry a version number and a "last updated" date on the page, and some are updated daily; a FRED series gains a row every release; so the file you have is one version of a moving thing. For datasets that are regularly updated, the date you downloaded it is what determines the date you should report.

2. Collector and purpose.

The page's description usually says where the numbers came from: a survey, an agency's program, a website that was scraped, an API that was queried, a paper. Whoever did that is the collector. Their purpose in collecting the dataset should be stated as the purpose of the dataset. On FRED and the BLS the collector is named on the page; on Kaggle it is often two links away. Where the description says nothing, the field says so. Some Kaggle pages say, a few paragraphs down, that the dataset is synthetic (generated by a program to look like data). That is the answer to field 3. Ideally, you should try to find a dataset that isn't synthetic: It is not appropriate to write that your dataset is synthetic as a limitation of your report, as you picked the dataset.

3. Collection.

"One row is one ..." can be one of the harder sentences in the dataset card to answer. Sometimes a row is a pairing (a tennis match has two players, and each appears in either column), a repeat (a song on a chart file once for every week it charted; a country in a country-by-year file once per year), or a snapshot (a listing as it stood on the day of the scrape). Say which, and say when the data were gathered.

4. Licence and permissions.

Kaggle prints the licence in the panel on the right of every dataset page, and it reads CCo (no rights reserved) on some, a Creative Commons attribution licence on others, "Other (specified in description)" on a few, and "Unknown" on many. "Unknown" is an answer to write down: the poster has not said what you may do with the file. Government data (FRED's sources, the BLS, data.gov) is usually public domain, and the page says so; a deposited research dataset (OSF) carries whatever licence the depositor chose. If you use a dataset collected internally by a researcher in the university, the dataset may not yet be licensed. If that is the case, ensure that this is properly labelled.

5. Structure.

Rows and columns, and the variable table. A file with 210 columns does not get a 210-row table. The table takes the columns the question might use (Part 4 narrows it further), and one sentence covers the rest (for example, "the remaining 190 columns are orbital and stellar measurements and their uncertainties"). "Stored as" is what `read_csv()` reported (dbl, chr, lgl, date), and it guesses from the first thousand rows: a price written \$1,299 might be read in as chr, and a code column of 1s and 2s might be represented as dbl.

6. Missing values.

Use `colSums()` of `is.na()` (or Python equivalent) first, as in Lab 1. Then `summary()` on every numeric column, looking for a value that cannot be true: -1, 0 where 0 is

impossible, 99, 999, 9999, a year of 1900. One Kaggle tennis file has no NA in any column and 15,652 rows whose ranking points are -1; `is.na()` does not detect those values yet you should report them as missing in your card.

Data may be disguised as text: `read_csv()` reads an empty cell and the letters NA as missing, and anything else (N/A, none, ?, -) arrives as text, so a column that should be numeric and is chr is the sign to look for.

A numeric column that arrived as chr is a finding for field 7. Converting it is a change for field 8, and the code for the conversion goes in the chunk there.

7. Known issues.

Check the page's Discussion tab, where available, for what earlier users report. Then write your own issues in R (or Python): duplicated rows (`duplicated()` returns TRUE for a row that repeats an earlier one, so the count is `sum(duplicated(d))`); an unnamed first column that an exported index left behind (`read_csv()` calls it `..1`); numbers stored as text; a column that takes one value in every row; dates stored as text in a format R/Python did not recognise.

8. Changes you made.

For this lab, probably none, or `na_if()` for a sentinel value. The code goes in the chunk and the sentence goes in the field.

9. Citation.

Some pages say how to cite (FRED and OSF do). Otherwise: the poster or agency (year). Title of the dataset. The site. The URL, and the date you accessed it. If field 2 named an original source, cite that as well.

Numbers in sentences, not printouts

In a report the Data section prints nothing (Lab 1, Part 2: no `str()`, no `summary()`). The sheet's looking chunk is set `include = FALSE`, so it runs and shows nothing, and the numbers go into sentences with inline R/Python: a backtick, the letter r (or the word python), a space, an expression, and a closing backtick. Written in the text, ``r nrow(d)`` is replaced by the number of rows when the sheet is knitted, and if the data change, the number changes with them. Two things to know: the expression has to stay on one line (a line break inside the backticks stops it being evaluated, and the reader sees the code instead of the number), and it is for numbers you would otherwise type by hand.

TASK 2 — COMPLETE THE CARD

Part 2 of the starter sheet is the nine fields with the looking chunk handed to you commented out. Please read the dataset's page, run the chunk, and fill every field. Where the page does not answer a field, say so in the field: "the page does not say who collected the data" is an answer but a blank is not.

Part 3: The research question

The research question is what every later section of a report answers, so the variables, the figures, and the discussion all follow from it. Four things make one usable.

It ends in a question mark.

A statement is not a question, and a question that is really a plan (“I will look at phone use and grades”) is not one either.

It has two or three sentences of background.

Why the question is interesting, and what a reader would learn from an answer. This is not a literature review; it is the bridge from the data to the question, and it sets up one half of the report’s arc. The discussion at the end is the other half.

It is specific.

The question should name the variables in their units, and which comparison or relationship your report needs to investigate. Please explore the data enough to know the units before writing the question.

A picture can answer it.

For this report’s theme, that means a question about how a quantity is distributed and whether the shape differs between groups, about how two quantities move together, or about how something changed over time. Two kinds of question do not work. “Does X cause Y?” cannot, in general, be answered from observed datasets (an observational dataset), so ask about the relationship instead. And “what is the average X?” is answered by one number, which is a statistic rather than a question.

NOTE — A WEAK QUESTION, AND A BETTER ONE

On PluckData (the course’s survey, which is an example here and not a dataset you may use), a weak question is: *Does phone use affect grades?* It asks for a cause but neither “phone use” nor “grades” is a column in the dataset, and there are no units which can be used to frame the question.

A better one:

Smartphone use is often blamed for falling grades, and the 121 students in this survey were measured on both. Among them, how does problematic phone use (MPPUS_Total, a score from 0 to 270) relate to grade point average (GPA, from 0 to 4), and does the pattern differ between men and women?

The hypotheses, in Lab 1’s form, if you want them:

H_0 : Phone use and GPA are unrelated

H_A : Higher phone use is associated with a lower overall GPA on average

TASK 3 — WRITE THE QUESTION

Part 3 of the sheet has a place for the background and one for the question or questions. Please write two or three sentences of background, then the question. Check your question against the four criteria above.

Please have the question set before you draw anything. A question written after seeing the pattern is HARKing (Lab 1, Part 2), and it is easy to do by accident in an exploratory report.

Part 4: The variables of interest

The variables of interest are the subset of field 5 that the question uses, and only those. Each gets one entry: the column name in code font, what it represents, its type (binary, nominal, ordinal, or continuous, and what it is stored as), its units, its range or its levels, and, if it helps, a center (the median or the mean). The reader has not seen the file, so GPA is spelt out as a “grade point average” on a 0–4 scale, and a column called grade says whether it is a letter grade or a year in school.

Format the numbers

The temptation is to print `summary()` and let the reader find the numbers. Please do not. Formatted, the three variables of the question above read:

- `MPPUS_Total` (continuous, stored `dbl`): the Mobile Phone Problem Use Scale total, 27 items each scored 0 to 10 and summed, so 0 to 270; observed 13 to 190, median 95.
- `GPA` (continuous, stored `dbl`): grade point average on a 0–4 scale; observed 2.03 to 4.00 once three values of 999 were treated as missing, median 3.34.
- `Sex` (binary, stored `dbl` in the CSV as 1 and 2; the `.rds` labels 1 as male and 2 as female): 51 men, 70 women, one missing.

Each number in those lines can be inline R, so that it is read from the data rather than typed. Turning `summary()` output into a list like this is also a reasonable use of generative AI: give it the output, the page’s documentation, and one finished line, and then check every number it returns against the output, because a model rounds and sometimes invents. Disclose it at the end of the sheet, as the syllabus asks.

TASK 4 — LIST THE VARIABLES

Part 4 of the sheet is a list with one entry per variable. Please fill it for the variables your question uses, in the format above, with the numbers computed rather than typed where you can.

When you have finished

Knit the sheet and upload both files and the data to Brightspace by Sunday night. Lab 1, Part 2, gives you a sense of how the analysis report starts: the Data section, the research question, and the variables of interest are its first three sections, and this lab's four parts are those three sections in draft, for a dataset you found. If it turns out to be the dataset for Report 1, then your draft here is a head start; if not, you have done the finding once and know what it takes.

READING — REFERENCE

- Lab 1, Part 3, for the nine fields and the four R functions the penguins card used – `unique()`, `table()`, `is.na()`, `colSums()` – or their Python equivalents.
- The Data Resources on Brightspace for more places to find data.
- Inline code in R Markdown/Quarto, on the cheatsheet: <https://rstudio.github.io/cheatsheets/rmarkdown.pdf>

SUBMISSION

- ☐ Knit your document to PDF
- ☐ Check that all code is visible (not running off the page)
- ☐ Check that all output displays correctly
- ☐ Submit both .Rmd/.qmd and .pdf files